Supaporn **Klabklaydee**

DOCTORAL STUDENT (2ND YEAR) · CHEMOINFORMATICS & ENVIRONMENTAL AI RESEARCHER

1-1-24, Minami-ku, Yokohama, Japan

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"Decoding Environmental Complexity Through AI-Driven Molecular Systems"

Summary

Doctoral researcher at the Institute of Science Tokyo specialising in **AI-driven molecular discovery for environmental and marine science**. I build end-to-end deep learning systems — spanning generative molecular modelling (VAE/GAT/GRU-SELFIES), graph neural networks, physics-informed neural networks, and large-scale protein language model embeddings — to decode the chemical complexity of dissolved organic matter (DOM) and marine enzyme diversity. My flagship project, the *CRAM Digital Twin*, integrates FT-ICR MS data with multi-task AI to simulate biotransformation pathways, predict toxicity, and model carbon cycling dynamics across the global ocean. Under my doctoral thesis with **Prof. Manabu Fujii** (scholarship recipient), I work on the **AISpread1000** programme (¥5,000,000) — molecular generation of a million-scale diverse-DOM dataset via k-NN chemical-space learning. I collaborate remotely with **JAMSTEC** and **RIKEN** on marine bioinformatics and HPC-scale enzyme family discovery. I publish open-source, reproducible pipelines at the interface of cheminformatics, bioinformatics, and computational environmental engineering.

Research Interests

AI / Deep Learning: GNN, GAT, MPNN, Graph Transformer, VAE, GRU-SELFIES, Physics-Informed NN, ESM-2 Protein Language Model, SHAP, Integrated Gradients

Cheminformatics & Bioinformatics: Molecular representation learning, *de novo* molecular generation, biotransformation pathway prediction, large-scale sequence embedding, enzyme family discovery, FT-ICR MS analysis

Environmental & Marine Science: Dissolved organic matter (DOM) characterisation, marine carbon cycling, PFAS fate & transport, emerging contaminant toxicity, ocean-scale kinetic modelling

Biomedical AI: Vaccine antigen optimisation, molecular docking & drug–receptor selectivity, collagen scaffold engineering, multi-objective evolutionary algorithms

High-Performance Computing: Distributed GPU computing (SLURM, A100/H100/V100), scalable bioinformatics pipelines, containerised reproducible workflows (Singularity, Snakemake)

Research Highlights

- **Published** in *ACS ES&T Water* (Jan 2026): multigenerational toxicity prediction with GNN (**GSAT model**, R^2=0.907, 98.9% faster; 4.7M reaction steps across 12 pollutants).
- **AISpread1000** (¥5M, PI: Prof. M. Fujii): doctoral research on million-scale diverse-DOM molecular generation via k-NN chemical-space learning.
- **Preprint** on *ChemRxiv*: structural & kinetic analysis of carboxyl-metabolizing enzymes; active site architecture + environmental adaptation across marine carbon cycling (w/ N. Sudsa-Ard, M. Fujii).
- **CRAM generative model** (VAE + GAT + GRU-SELFIES): median candidate isomers per marine formula 1.4 $\rightarrow$ 9.3.
- **Physics-Informed Neural Networks** (R^2=0.93) for ocean-depth chemical degradation (0–6000 m); validated vs. ARGO/GLODAP (300K+ measurements).
- **Graph Transformer** for biotransformation pathway prediction: 210K+ reaction pairs, 87% Top-3 accuracy.
- **Marine enzyme discovery**: weighted dual-embedding (ESM-2 + k-mer, Mahalanobis) + hierarchical DBSCAN on 1.29M sequences → **847 families**, 156 novel, Silhouette 0.47.
- **Workshop instructor**: Cheminformatics (IST, Liverpool); **JAXA** graph theory for satellite & stellar data (3 classes); marine bioinformatics (2 classes).
- **Best Student Cohort**, JST × LLNL DSSI Summer Internship 2025.
- **MEXT Full Scholarship** (2024) for Master–PhD programme, Institute of Science Tokyo.

Education

Institute of Science Tokyo (formerly Tokyo Institute of Technology)
Tokyo, Japan

PH.D. IN CIVIL AND ENVIRONMENTAL ENGINEERING — *Sep. 2025 – Present*

- Dissertation: "Digital Twin of MS2 Fragmentation of Carboxyl-Rich Alicyclic/Aliphatic Matters (CRAMs): Formula Assignment (MS1), Biotransformation Bidirectional Model and Kinetic Model"
- Advised by Prof. Manabu Fujii

Institute of Science Tokyo
Tokyo, Japan

UNIVERSITY-WIDE PROGRAM IN DATA SCIENCE & AI — EXPERT LEVEL PLUS
2025

- Advanced certification in machine learning, deep learning, and AI applications for science and engineering.

Institute of Science Tokyo
Tokyo, Japan

M.ENG. IN CIVIL AND ENVIRONMENTAL ENGINEERING
Oct. 2023 – Sep. 2025

- Thesis: "Multigenerational Toxicity Prediction Using Graph Neural Networks and Biotransformation Analysis"
- GPA: 4.16 / 4.5 Advised by Prof. Manabu Fujii

King Mongkut's University of Technology Thonburi (KMUTT)
Bangkok, Thailand

B.ENG. IN ENVIRONMENTAL ENGINEERING (INTERNATIONAL PROGRAM), 1ST CLASS HONOURS, DEAN'S LIST
Jun. 2019 – May 2023

- **Best Rising Star Award** (University-wide, Physical Sciences) — recognised as the outstanding graduating student across all physical science undergraduate programmes.
- Thesis: "Molecularly Imprinted TiO_2 for Selective Removal of PPCPs in Wastewater of the Wooden Industry with Quantum-Mechanistic Calculation"
- GPA: 3.75 / 4.0 Advised by Assoc. Prof. P. Kemacheevakul, Assoc. Prof. S. Chuangchote, and Prof. T. Sagawa

Work & Internship Experience

JAMSTEC & RIKEN
Remote / Japan

REMOTE RESEARCH COLLABORATOR — MARINE BIOINFORMATICS & HPC COMPUTING
Dec. 2025 – Present

- Conducting ongoing remote collaboration with **JAMSTEC** (Japan Agency for Marine-Earth Science and Technology) and **RIKEN** on **large-scale marine metagenomics** and enzyme family discovery; utilising national HPC resources for distributed sequence embedding and clustering across global ocean datasets (TARA Oceans, OSD, GOS, Malaspina).
- Engineering **scalable dual-embedding pipelines**: ESM-2 protein language model (480-dim transformer embeddings, half-precision inference) fused with k-mer frequency vectors (8,000-dim, k=4) via **Mahalanobis-distance weighted combination**; implemented chunked batching and memory-mapped HDF5 I/O to process **1.29M+ sequences** without memory overflow on large-memory nodes.
- Designed and benchmarked **hierarchical DBSCAN** (HDBSCAN, `min_cluster_size`=15, `min_samples`=5) clustering workflow on HPC: SLURM job arrays for parameter sweeps, **GPU-accelerated approximate nearest-neighbour** (FAISS IVF-PQ index) to reduce pairwise distance computation from $O(n^2)$ to $O(n \log n)$; achieved full 1.29M-sequence clustering in <**6 h** wall time.
- Discovered **847 marine enzyme families** (78.3% EC-label consistency, Silhouette 0.47 vs. 0.32 baseline), including **156 novel putative families** with no known EC annotation; applied **UMAP** (cosine metric, $n\_neighbours$=30) for 2D/3D embedding visualisation stratified by ocean depth, temperature, and geographic zone.
- Built **environmental adaptation regression models** (XGBoost + SHAP) on cluster-level physicochemical summaries to identify sequence motifs predictive of thermal stability and substrate specificity across depth zones (0–6000 m); temperature–depth metadata explains **34.7%** of inter-family embedding variance (R^2).
- Developing **in-preparation manuscript**: *"Deep Learning-Guided Discovery of Marine Enzyme Families: A Weighted Dual-Embedding Approach for Environmental Adaptation Analysis"* (w/ N. Subsa-ard, K. Iwabuchi, J. Liu, M. Fujii).
- Maintaining **containerised, reproducible HPC workflows** (Singularity + Snakemake DAGs, Git-LFS dataset versioning) aligned with open-science data practices for the global marine enzyme atlas in collaboration with JAMSTEC and RIKEN computational groups.

Institute of Science Tokyo
Tokyo, Japan

WORKSHOP INSTRUCTOR — CHEMINFORMATICS (INTERNAL)
2025 – 2026

- **Class 1 — Molecular Representations & RDKit Foundations:** taught students to read and write **SMILES** and **SELFIES**, parse molecules in RDKit, compute basic physicochemical properties ($logP$, TPSA, molecular weight), and visualise 2D structures; participants built a mini pipeline from SMILES string → molecule object → property table.
- **Class 2 — Fingerprints, Similarity & Intro GNNs:** covered Morgan/ECFP fingerprints, Tanimoto similarity search, chemical-space visualisation (PCA/UMAP on fingerprint matrices), and an introductory **graph neural network** for molecular property prediction; students trained a small GNN baseline on a public MoleculeNet-style task and interpreted attention/message-passing intuition.

Japan Aerospace Exploration Agency (JAXA)
Japan

WORKSHOP INSTRUCTOR — GRAPH THEORY FOR SATELLITE & STELLAR DATA
2025 – 2026

- **Class 3 — Graph Theory Foundations for Satellite Datasets:** introduced graph terminology (nodes, edges, adjacency, degree, paths, connectivity) and how Earth-observation / satellite data can be cast as graphs — spatial grids, orbit-ground station links, and sensor co-occurrence networks; students constructed toy graphs from sample satellite metadata.
- **Class 4 — Graph Algorithms on Satellite Networks:** taught shortest paths, centrality, community detection, and graph Laplacians applied to satellite constellation and Earth-observation workflows; participants practised network analysis in Python (NetworkX) on satellite-link and spatial-neighbourhood graphs.
- **Class 5 — Stellar Project — Graph Methods for Stellar / Astronomical Data:** project-based class applying graph representations to stellar datasets (star catalogues as similarity graphs, clustering of stellar populations, visualisation of graph communities); students delivered a mini **Stellar Project** linking graph theory to astronomical data interpretation.

Marine Bioinformatics Workshop
Japan

WORKSHOP INSTRUCTOR — MARINE BIOINFORMATICS
2025 – 2026

- **Class 6 — Marine Metagenomics & Sequence Pipelines:** hands-on marine bioinformatics — FASTA handling of ocean-derived sequences, quality control, k-mer feature extraction, and interpretation of marine microbial / enzyme sequence diversity from environmental DNA datasets.
- **Class 7 — Marine Enzyme Discovery with ESM-2 & Clustering:** taught **ESM-2** embedding extraction for marine protein sequences, UMAP visualisation stratified by ocean context, and **hierarchical DBSCAN / HDBSCAN** clustering to discover enzyme families; students evaluated Silhouette scores and linked clusters to ecological or functional hypotheses.

University of Liverpool
Liverpool, UK / Remote

WORKSHOP INSTRUCTOR — CHEMINFORMATICS (EXTERNAL)
2025 – 2026

- **Class 8 — Applied Cheminformatics for Molecular Discovery:** delivered an external cheminformatics class covering end-to-end workflows — molecular sanitisation, descriptor engineering, QSAR-style supervised learning, and structure-aware model evaluation (scaffold split, leakage awareness); students practised building reproducible Jupyter notebooks for chemical property modelling relevant to drug discovery and environmental toxicology.

JST × Lawrence Livermore National Laboratory (LLNL) — Data Science Institute (DSI)
Livermore, CA, USA

HPC / AI RESEARCH INTERN (JST EXCHANGE)
Jul. 2025 – Sep. 2025

- Conducted research at a **Tier-1 DOE national HPC facility** (El Capitan / Lassen cluster, DOE NNSA); exploited **GPU-accelerated computing** (NVIDIA A100/H100) and multi-node parallelism via **SLURM** job arrays for large-scale evolutionary peptide search.
- Project: *"Structurally-Aware Genetic Algorithm for Vaccine Antigen Optimisation"* (w/ S. Conti, D. Faissol, F. L. da Silva, Global Security Directorate); designed and implemented a **structure-conditioned genetic algorithm (GA)** that encodes antigen peptide sequences as SMILES/SELFIES graphs and evaluates fitness via **ESM-2 protein language model** embeddings, optimising simultaneously for immunogenicity score, structural stability (AlphaFold2 pLDDT proxy), and sequence diversity.
- Engineered **distributed Python pipelines** (PyTorch, `torch.multiprocessing`, Ray) to parallelise population-based crossover/mutation across 8× A100 nodes; reduced per-generation wall time from **4.2 h** → **18 min** (14× speed-up) through vectorised SELFIES batch inference and GPU-resident fitness evaluation.
- Implemented **multi-objective Pareto-front selection** (NSGA-II variant) balancing three competing objectives; applied **active learning** with a surrogate Gaussian Process to propose high-utility mutation operators, cutting wet-lab candidate shortlist from 10,000 → 47 sequences while maintaining hypervolume coverage.
- Conducted **ablation studies** and interpretability analysis: visualised ESM-2 attention weights with **BERTViz** and attributed fitness scores to residue positions via **Integrated Gradients**, identifying key immunogenic epitope regions consistent with known MHC-II binding motifs.
- Delivered **reproducible HPC workflow** (Snakemake + Singularity containers) enabling full experiment replay from raw FASTA input to ranked antigen library; workflow archived in LLNL internal GitLab.
- Awarded **Top Presenter** at the DSSI Summer Case Competition and **Best Student Cohort** among all JST × LLNL DSSI interns (2025 cohort).

Research Affairs, Faculty of Medicine, Chulalongkorn University
Bangkok, Thailand

COMPUTATIONAL RESEARCH ASSISTANT
Mar. 2022 – Present

- Performed computational analyses for the **NRCT-funded** project: *"Collagen Scaffold with Gallic Acid-Assisted EDC/NHS Crosslinking: Potential Application for Bone Tissue Engineering"* — contributed molecular-level mechanistic understanding to an experimental biomaterials team.
- Executed **molecular docking campaigns** (AutoDock Vina, SMINA/Vinardo) to characterise non-covalent pre-reaction binding of crosslinking agents (EDC, NHS, gallic acid derivatives) to collagen triple-helix and MMP-1 receptor sites; processed and annotated 2,000+ docking poses.
- Conducted **MD simulations** (GROMACS, CHARMM36 force field, 100 ns) of collagen–crosslinker complexes; computed binding free energies (MM-PBSA), RMSD stability metrics, and hydrogen-bond occupancies to rank ligand efficacy.
- Built a **heterogeneous GNN** (PyTorch Geometric) to learn ligand–protein interaction features directly from docking graphs; multi-task Model D achieved validation RMSE **0.55 kcal/mol** and Spearman ρ=**0.95** for Collagen Selectivity Index (CSI) prediction.
- Applied **Integrated Gradients** and attention visualisation to confirm GNN pharmacophore recovery; identified pyrogallol hydroxyl groups and galloyl ester linkages as key selectivity determinants; manuscript in preparation (w/ N. Darumas, P. Kueanjinda et al.).

King Mongkut's University of Technology Thonburi (KMUTT)
Bangkok, Thailand

TEACHING ASSISTANT — UNDERGRADUATE LABORATORY COURSES
2022 – 2023

- **EN303 Dynamical Systems Colloquium Laboratory** — supervised 30+ undergraduates in weekly lab sessions covering dynamical systems modelling, numerical integration, and simulation; graded reports and provided individual feedback; collaborated with Kyoto University (Quantum Energy Processes Laboratory exchange).
- **EN321 Quantum Energy Processes Laboratory** — assisted instruction in quantum energy processes experiments (photovoltaic characterisation, energy storage); demonstrated equipment operation, guided data acquisition and uncertainty analysis; supported Kyoto University academic exchange programme.
- Developed supplementary **Python-based data analysis scripts** for students to visualise experimental results (time-series, Bode plots, phase portraits); introduced version-control best practices (Git) to undergraduate cohort.
- Held weekly office hours; mentored 5 students who subsequently pursued research internships and graduate study.

Kyoto University, Laboratory of Prof. Takashi Sagawa (Quantum Energy Processes) *Kyoto, Japan*

RESEARCH STUDENT (EXCHANGE INTERN) *Jun. 2022 – Oct. 2022*

- Secured competitive research funding from **AMGEN Japan (2022)** to investigate **photocatalytic degradation of pharmaceuticals and personal care products (PPCPs)** in wastewater; conducted UV/solar-driven oxidation experiments using modified TiO_2 and ZnO photocatalysts under controlled reactor conditions.
- Performed **kinetic modelling** of PPCP degradation: fitted Langmuir–Hinshelwood rate equations to time-resolved HPLC/UV-Vis data; identified pseudo-first-order rate constants (k_{obs}) and correlated them with photocatalyst surface area and reactive-oxygen-species (ROS) yield.
- Ran **DFT calculations** (Gaussian, B3LYP/6-31G*) on selected PPCP molecules to compute electron affinity, HOMO–LUMO gaps, and radical-attack susceptibility sites; used computed descriptors as features for a **Gaussian Process regression** model predicting k_{obs} from molecular structure alone (leave-one-out RMSE: 0.031 min^{-1}).
- Presented findings at the **Kyoto International Symposium** and the **Virtual Amgen Asia Symposium**; received commendation for clarity of computational analysis in an experimental research group.
- Gained hands-on exposure to Japanese academic laboratory culture, scientific communication in English/Japanese, and cross-cultural research collaboration within a quantum-energy-processes research group.

Ministry of Social Development and Human Security of Thailand (MSDHS) *Bangkok, Thailand*

ENVIRONMENTAL ENTERPRISING PROMOTER *Jan. 2020 – Jul. 2020*

- Led national campaign aligning environmental activities with **UN Sustainable Development Goals (SDGs)**; designed data-driven outreach strategies to quantify community engagement and behavioural change.
- Applied **NLP-based text analysis** to synthesise policy documents and public feedback, extracting key sustainability indicators to inform programme prioritisation.
- Developed **statistical models** (regression, clustering) to segment community groups by environmental awareness level and predict uptake of sustainable practices, supporting evidence-based advocacy.
- Coordinated cross-sector partnerships with government agencies, NGOs, and universities to scale environmental education initiatives across 5+ provinces.

United Nations (Online Session) *Remote*

ENVIRONMENTAL INTERN *Feb. 2019 – May 2019*

- Contributed to UN online sessions on **environmental climate change and energy conservation**; synthesised multi-source environmental datasets and applied **regression and time-series analysis** to quantify trends in greenhouse-gas emissions and energy-use efficiency across member states.
- Prototyped a **supervised ML pipeline** (Random Forest + gradient boosting) to classify countries by climate-vulnerability index using socioeconomic and geospatial indicators; achieved 82% cross-validated accuracy, surfacing top predictive features for policymaker briefings.
- Explored **LSTM-based sequence models** to forecast short-term CO_2 concentration trajectories from NOAA atmospheric records; demonstrated 15% lower MAE vs. ARIMA baseline over a 12-month horizon.
- Prepared data visualisation dashboards (Matplotlib, Seaborn) and contributed to a **scenario-modelling report** on net-zero transition pathways; findings presented to a panel of international policymakers and environmental engineers.

King Mongkut's University of Technology Thonburi (KMUTT), Secondary Science Division *Bangkok, Thailand*

INDEPENDENT RESEARCH STUDENT — ENVIRONMENTAL SCIENCE & INVENTION *2017 – 2019*

- Designed and conducted an **independent research project** on low-cost environmental sensing and remediation; developed a prototype sensor system for monitoring water-quality indicators (pH, turbidity, dissolved oxygen) interfaced with a **microcontroller (Arduino)** and custom data-logging firmware.
- Applied **statistical data analysis** (Excel, early Python scripting) to multi-site water sampling datasets; identified pollution hotspots using correlation analysis between industrial activity and sensor readings.
- Awarded **Gold Award at I-Envex IYIA 2018** (International Young Inventors Award) and selected as **Thailand Representative to Intel ISEF 2019** (Phoenix, AZ, USA) — competed among 1,800+ projects from 80+ countries.
- Represented Thailand at **WTUN Hackathon for Climate Action 2020** (2nd Runner-up) and **NRCT Thailand New Gen Award 2020** (2nd Runner-up & Gold Medal), continuing independent environmental research through undergraduate entry.

Research Experience

Fujii Laboratory, Institute of Science Tokyo *Tokyo, Japan*

Research Assistant / Doctoral Researcher *Oct. 2023 – Present*

- **ACS ES&T Water** (Jan 2026): "Multigenerational Toxicity Prediction Using GNN" — GSAT model, R^2=0.907, **98.9% faster** computation; 4.7M reaction steps across 12 pollutants.
- **AISpread1000** (¥5M, PI: Prof. Manabu Fujii): doctoral thesis work on molecular generation for a **million-scale diverse-DOM** dataset via k-NN projection of chemical-space learning — expanding generative coverage of dissolved organic matter structural diversity.
- **ChemRxiv preprint**: "Structural and Kinetic Analysis of Carboxyl-Metabolizing Enzymes" — FPocket analysis, MD simulations, active site architecture across marine carbon cycling (w/ N. Subsa-ard, K. Iwabuchi).
- **CRAM generative model**: property-conditioned VAE + GAT encoder + GRU-SELFIES decoder; median candidate isomers per formula **1.4** $\rightarrow$ **9.3**.
- **Quantum fidelity kernel benchmark** (null-result study): scaffold-split learning-curve comparison of an 8-qubit PennyLane fidelity-kernel SVM vs. Random Forest, RBF-SVM, and AttentiveFP on MoleculeNet (BBBP, BACE, HIV, Tox21); 280 model–condition pairs; no global quantum advantage over best classical baselines.
- **Hybrid GAT + MPNN** for molecular half-life prediction (R^2=0.847) with SHAP/Integrated Gradients; Gradio web interface for real-time predictions.
- **Marine Enzyme Discovery**: weighted dual-embedding (ESM-2 480-dim + k-mer 8,000-dim, Mahalanobis) + hierarchical DBSCAN on **1,293,763 sequences**; **847 families** (78.3% EC consistency), 156 novel putative families; Silhouette 0.47 vs. 0.32; temperature–depth explains 34.7% variance.
- **Physics-Informed NN** (R^2=0.93) for chemical degradation 0–6000 m; **17,500**$\times$ **half-life variation** discovered; validated vs. ARGO/GLODAP (300K+ measurements).
- **Graph Transformer** for biotransformation pathway prediction: 210K+ reaction pairs, 99 classes, **87% Top-3 accuracy**; Focal Loss + SMILES augmentation + SWA.
- JST topic: **PFAS** transformation product identification via mass-difference network — *Water Research* doi:10.1016/j.watres.2025.123645.
- JST topic: **Chemical Space Refinement Framework** for DOM structural complexity — ChemRxiv doi:10.26434/chemrxiv-2025-9s13m-v2.
- **MOSAIC-GNN** (in prep.): multi-graph neural network for high-accuracy DOM **molecular formula assignment** from MS1-only FT-ICR MS data; designed heterogeneous graph construction encoding mass differences, isotope patterns, and elemental constraints; targeting sub-ppm formula prediction without MS/MS fragmentation.
- **Divergent CRAM Formation** (in prep.): AI-expanded MS/MS spectral libraries via GRU-SELFIES generation to characterise CRAM formation and persistence divergence across **freshwater vs. marine** environments; integrated FT-ICR MS fingerprinting with transformer-based fragment prediction.
- **Digital Twin — Kinetic Model** (in prep.): physics-constrained deep learning model for emerging-chemical fate kinetics at **global ocean scale**; coupled advection-diffusion PDEs with neural ODE solver to simulate depth- and latitude-resolved half-life distributions.
- **Breaking the CRAM Behavior** (in prep.): generative AI pipeline (VAE + GAT) to design novel molecules that **unlock hidden enzymatic pathways** in marine carbon cycling; combined *de novo* molecular generation with enzyme-docking scoring (AutoDock Vina) and ESM-2-based substrate-specificity filtering.

Research Affairs, Faculty of Medicine, Chulalongkorn University (NRCT-funded) *Bangkok, Thailand*

Computational Research Assistant *Mar. 2022 – Present*

- Project: "Collagen Scaffold with Gallic Acid-Assisted EDC/NHS Crosslinking for Bone Tissue Engineering" — conducted **2,052 SMINA/Vinardo rigid-receptor docking** calculations for 9 ligands across 76 collagen binding sites at 3 pH values, plus 120 MMP-1 cross-receptor runs.
- **Heterogeneous GNN framework** learning ligand–protein interaction features from docking graphs; **Model D (multi-task)** achieved validation RMSE **0.55 kcal/mol** and Spearman ρ=**0.95** between predicted and Vinardo-derived Collagen Selectivity Index (CSI).
- Ligand identity explained **74.1%** of binding-energy variance (η^2); pH had no significant effect; PGG and ellagic acid identified as collagen-selective leads (CSI $\leq$ 0.97).
- **Integrated Gradients** interpretation confirmed GNN recovers chemically meaningful pharmacophore patterns — pyrogallol hydroxyl groups in compact phenols, galloyl ester linkages in PGG.
- Sensitivity analysis at exhaustiveness = 64 confirmed PGG ranking stability; manuscript in preparation (w/ N. Darumas, P. Kueanjinda, N. Subsa-ard, N. Charuvajana, A. Uppakarnsang).

Research Center of Advanced Materials for Energy and Environmental Technology (MEET), KMUTT — *Bangkok, Thailand*

Research Student — *Oct. 2022 – Sep. 2023*

- Undergraduate thesis: *"Molecular Imprinted TiO_2 for Organic Pollutants Removal in Wastewater"* — synthesised **molecularly imprinted TiO_2 nanoparticles** via sol-gel method; systematically optimised calcination temperature (400–700 °C), imprinting ratio, and pH to maximise photocatalytic degradation of phenol under UV irradiation.
- **DFT simulations** (Gaussian 16, B3LYP/6-31G* and STO-3G basis sets): computed HOMO–LUMO gaps, electron density maps, and adsorption binding energies for phenol, bisphenol-A, and template molecules on TiO_2 surface clusters; correlated electronic descriptors (TPSA, ionisation potential, aromaticity) with experimentally measured degradation rates ($k_{\text{obs}}, r^2 > 0.88$).
- **Molecular dynamics (MD) simulations** (GROMACS, OPLS-AA force field): modelled pollutant–MOF host–guest interactions over 50 ns trajectories; computed RMSD, RMSF, and radial distribution functions to characterise selectivity and diffusion behaviour of target pollutants within MOF pores.
- Trained a **Random Forest regression model** (scikit-learn) on a compiled dataset of 320 pollutant–photocatalyst pairs (molecular fingerprints + DFT descriptors) to predict pseudo-first-order degradation rate constants; achieved **RMSE 0.042 min^{-1}** (5-fold CV), identifying functional-group importance via SHAP values.
- Studied **Metal–Organic Frameworks (MOFs)** with molecular imprinting for selective industrial pollutant removal; characterised pore geometry (BET surface area, pore-size distribution) and matched selectivity to DFT-predicted host–guest binding energies.
- Applied **interpretable machine learning**: trained Gradient Boosting and Random Forest regressors on DFT + molecular descriptor features; used **SHAP summary plots**, partial dependence plots, and **Integrated Gradients** to identify the top physicochemical drivers of photocatalytic degradation efficiency, revealing that HOMO energy and hydroxyl-group count collectively explain **61%** of k_{obs} variance.

Publications

Underline = Supaporn Klabklaydee (first/co-first author). † = preprint.

Journal Articles

1. S. Klabklaydee, B. Zhang, J. Liu, …, and M. Fujii, "Multigenerational Toxicity Prediction Using Graph Neural Networks and Biotransformation Analysis," *ACS ES&T Water*, Jan. 2026.
2. B. Zhang, J. Liu, S. Qing, …, and M. Fujii, "Accurate Detection and High Throughput Profiling of Unknown PFAS Transformation Products for Elucidating Degradation Pathways," *Water Research*, Apr. 2025.

Preprints

1. S. Klabklaydee, N. Sudsa-Ard, and M. Fujii, "Structural and Kinetic Analysis of Carboxyl-Metabolizing Enzymes: Understanding Active Site Architecture and Environmental Adaptation in Marine Carbon Cycling," *ChemRxiv*, 2025. †
2. J. Liu, R. Gao, M. Elsamadony, …, and M. Fujii, "A Chemical Space Refinement Framework for Decoding the Structural Complexity of Dissolved Organic Matter," *ChemRxiv*, 2025. †
3. S. Klabklaydee and M. Fujii, "MOSAIC-GNN: A Multi-Graph Neural Network for High-Accuracy Dissolved Organic Matter Molecular Formula Assignment from MS1-Only High-Resolution Mass Spectrometry," in preparation.
4. S. Klabklaydee and M. Fujii, "Divergent CRAM Formation and Persistence Across Freshwater and Marine Environments Revealed by AI-Expanded MS/MS Libraries," in preparation.
5. S. Klabklaydee and M. Fujii, "Digital Twin of Kinetic Modelling of Emerging Chemical in Global Scale Marine System," in preparation.
6. S. Klabklaydee and M. Fujii, "Breaking the CRAM Behavior: AI-Generated Molecules Unlock Hidden Enzymatic Pathways in Marine Carbon Cycling," in preparation.
7. S. Klabklaydee, N. Subsa-ard, K. Iwabuchi, J. Liu, and M. Fujii, "Deep Learning-Guided Discovery of Marine Enzyme Families: A Weighted Dual-Embedding Approach for Environmental Adaptation Analysis," in preparation.
8. N. Darumas, S. Klabklaydee, P. Kueanjinda, N. Subsa-ard, N. Charuvajana, and A. Uppakarnsang, "Graph Neural Network–Guided Molecular Docking of Nine EDC/NHS Collagen-Crosslinking Agents Reveals pH-Independent Affinity Hierarchy and Dual-Receptor Selectivity Over Porcine MMP-1," in preparation.

Technical Reports & Posters

1. S. Klabklaydee, S. Conti, D. Faissol, and F. L. da Silva, "Structurally-Aware Genetic Algorithm for Vaccine Development," *LLNL Data Science Summer Institute*, Livermore, CA, Sep. 2025. [Poster PDF]

2. S. Klabklaydee and M. Fujii, "Breaking the CRAM Behavior: AI-Generated Molecules Unlock Hidden Enzymatic Pathways in Marine Carbon Cycling," *NAIST Data Science Center Symposium*, Sep. 2025. [Abstract PDF]
3. S. Klabklaydee, "ToxiGraph: Predictive Toxicity Modeling with Molecular Descriptors and Graph-Based Analysis for LC50," *Water and Environment Technology Conference (WET)*, Japan, 2024. [Conf. ID: 00030901]

Honors & Awards

International Awards & Competitions

2025	**Best Student Cohort / Top Presenter,** DSSI Summer Internship, JST × Lawrence Livermore National Laboratory	*Livermore, CA, USA*
2020	**2nd Runner-up & Gold Medal,** NRCT Thailand New Gen Award, National Research Council of Thailand	*Thailand*
2020	**2nd Runner-up (Thailand Representative),** WTUN Hackathon for Climate Action	*Thailand*
2020	**2nd Runner-up (Thailand Representative),** International Poster Presentation Competition (IPPC) 2020 — Group A, Applied Sciences & Engineering	*International*
2019	**Participant (Thailand Representative),** Intel International Science and Engineering Fair (Intel ISEF)	*Phoenix, AZ, USA*
2018	**Gold Award,** I-Envex IYIA 2018 — International Young Inventors Award	*International*

Scholarships & Academic Honours

2026	**Doctoral Researcher — AISpread1000 Programme (¥5,000,000),** PI: Prof. Manabu Fujii · Million-scale diverse-DOM generation via k-NN chemical-space learning	*Institute of Science Tokyo*
2025	**Expert Level Plus Certification,** Data Science & AI Programme, Institute of Science Tokyo	*Tokyo, Japan*
2024	**MEXT Japanese Government Scholarship (Full scholarship + stipend, Master–PhD),** Ministry of Education, Culture, Sports, Science and Technology, Japan	*Tokyo, Japan*
2023	**Best Rising Star Award,** Physical Sciences Undergraduate Research, King Mongkut's University of Technology Thonburi	*Bangkok, Thailand*
2023	**1st Class Honours with Dean's List,** B.Eng. in Environmental Engineering, King Mongkutś University of Technology Thonburi	*Bangkok, Thailand*
2022	**AMGEN Japan Research Funding (2022),** Kyoto University Exchange Research Programme	*Kyoto, Japan*
2019	**Full Scholarship + Stipend (Bachelor),** King Mongkutś University of Technology Thonburi	*Bangkok, Thailand*
2016	**Full Scholarship + Stipend (High School),** Thai Government Scholarship	*Thailand*

Skills

Programming	Python, R, Julia, Java, C++, C
ML / DL	PyTorch, PyTorch Geometric, Hugging Face (ESM-2), Scikit-learn
Cheminformatics	RDKit, DeepChem, SELFIES, OpenBabel
DL Methods	GNN, VAE, Physics-Informed NN, Graph Transformer, MPNN
XAI	SHAP, Integrated Gradients, Captum
Deployment	Gradio, Docker, AWS / GCP (spot instances)
HPC / Infra	SLURM, multi-node GPU (A100/H100), MPI, parallel Python (multiprocessing, Ray)
Simulation	GROMACS (MD), Gaussian (DFT B3LYP), AutoDock Vina
Mass Spec.	FT-ICR MS, MZmine, Molecular Formula Assignment
Data / Stats	DBSCAN, UMAP, Pandas, NumPy, Matplotlib
Domains	Chemoinformatics, Marine Carbon Cycling, PFAS, Environmental Fate
Languages	Thai (native), English (professional), Japanese (intermediate)